Milestone 1 (Draft)

CSC 676 - Spring 2026

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Version History:

Version	Description	Date
v0.1	Initial draft for in-class review; sample entities and requirements.	2026-02-08
v0.2	Update for draft #2; Functional Database complete, conceptual design begun, Entity Set Desc. Begun	2026-02-17

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Project Description

- **High-Level Overview:**
    - The Transit Incident & Maintenance Reporting System improves how passengers experience and interact with local rail platforms like BART and Muni. Unlike existing apps that focus on static schedules and basic navigation, our platform delivers live, actionable data: real-time train crowding, station congestion, accurate next-train predictions, and immediate incident reporting for accidents or hazards. With this system, passengers get the information they need to make smarter travel decisions, while transit operators and first responders gain a unified view of system-wide conditions to dispatch resources more effectively. A public transparency portal keeps riders informed of active issues and resolutions, reducing duplicate reports and building trust in the system.

- **Unique Features:**
    - Recursive Delay Propagation: Use recursive relationships to automatically calculate the “downstream” impact of a single train delay on every station following it on the line.
    - Community Transparency Portal: To be more efficient than existing platforms like the “BART Watch” App. To improve the passenger experience, our system includes a public-facing communication space. By having this, it is easier for the community to see follow-up notes on active messes or delays. It prevents the issue of duplicate reporting by allowing users to see if an issue is already 'In-Progress' or 'Resolved' in real-time.

Functional Database Requirements

1. **User**
    - 1.1. A user shall have exactly one unique user ID.
    - 1.2. A user shall have exactly one unique username.
    - 1.3. A user shall have exactly one password.
    - 1.4. A user shall have exactly one unique email address.
    - 1.5. A user shall submit zero or many reports.

2. **Admin**
    - 2.1. An admin shall reference exactly one user ID.
    - 2.2. An admin shall have exactly one unique admin ID.
    - 2.3. An admin shall have exactly one role name.
    - 2.4. An admin shall have exactly one access level.
    - 2.5. An admin shall have exactly one admin status.
    - 2.6. An admin shall manage system data, including stations, lines, and agencies.

3. **Employee**
    - 3.1. An employee shall reference exactly one user ID.
    - 3.2. An employee shall have exactly one unique employee ID.
    - 3.3. An employee shall have exactly one job title.
    - 3.4. An employee shall have exactly one employment status.
    - 3.5. An employee shall have exactly one hire date.
    - 3.6. An employee shall be assigned to exactly one station.

4. **Officer**
    - 4.1. An officer shall reference exactly one user ID.
    - 4.2. An officer shall have exactly one unique officer ID.
    - 4.3. An officer shall have exactly one unique badge number.
    - 4.4. An officer shall have exactly one rank.
    - 4.5. An officer shall have exactly one assigned shift.

5. **Agency**

5.1. An agency shall have exactly one unique agency ID.

5.2. An agency shall have exactly one unique name.

5.3. An agency shall have exactly one headquarters location.

5.4. An agency shall have exactly one contact phone number.

5.5. An agency shall operate zero or many lines.

5.6. An agency shall operate zero or more vehicles.

5.7. An agency shall serve zero or many stations through Station_Agency.

6. Station

6.1. A station shall have exactly one unique station ID.

6.2. A station shall have exactly one name.

6.3. A station shall have exactly one city.

6.4. A station shall have exactly one zone code.

6.5. A station shall be served by many agencies through Station_Agency.

6.6. A station shall belong to zero or many lines.

6.7. A station shall be assigned to zero or more employees.

6.8. A station shall be associated with zero or many lines through Line_Station.

7. Station_Agency

7.1. A Station_Agency shall reference exactly one station ID.

7.2. A Station_Agency shall reference exactly one agency ID.

7.3. A Station_Agency shall store exactly one service start date.

7.4. A Station_Agency shall store zero or one service end date.

7.5. A Station_Agency shall store zero or one service note.

7.6. A Station_Agency (station ID, agency ID) pair shall be unique.

8. Mode (of transport)

8.1. A mode shall have exactly one unique mode ID.

8.2. A mode shall have exactly one unique name.

8.3. A mode shall have exactly one description.

8.4. A mode shall have exactly one typical use case.

8.5. A mode shall be assigned to zero or many lines.

9. Line

9.1. A line shall have exactly one unique line ID.

9.2. A line shall have exactly one unique name.

9.3. A line shall have exactly one line color.

9.4. A line shall have exactly one service pattern.

9.5. A line shall belong to exactly one agency.

9.6. A line shall be assigned to exactly one mode.

9.7. A line shall be associated with zero or many stations through Line_Station.

9.8. A line shall be assigned to zero or many trips.

10. Line_Station

10.1. A Line_Station shall reference exactly one line ID.

10.2. A Line_Station shall reference exactly one station ID.

10.3. A Line_Station shall store exactly one sequence number.

10.4. A Line_Station shall store zero or one platform information value.

10.5. A Line_Station shall store exactly one direction label.

10.6. A Line_Station (line ID, station ID) pair shall be unique.

10.7. A Line_Station (line ID, sequence number) pair shall be unique.

11. Incident_Report

11.1. An Incident_Report shall have exactly one unique incident ID.

11.2. An Incident_Report shall be reported by exactly one user.

11.3. An Incident_Report shall store exactly one description.

11.4. An Incident_Report shall store exactly one reported timestamp.

11.5. An Incident_Report shall store exactly one incident source.

11.6. An Incident_Report shall be assigned to zero or many employees through Dispatch_Assignment.

11.7. An Incident_Report shall have zero or many Incident_Photo.

11.8. An Incident_Report shall belong to zero or many stations through Incident_Station.

11.9. An Incident_Report shall belong to zero or many trains through Incident_Train.

11.10. An Incident_Report shall reference exactly zero or one Incident_Status.

11.11. An Incident_Report shall reference exactly one Report_Category.

11.12. An Incident_Report shall reference exactly one weather status.

11.13. An Incident_Report shall reference exactly one priority level.

11.14. An Incident_Report shall reference zero or many hazards.

12. Incident_Station

12.1. An Incident_Station shall reference exactly one Incident_Report ID.

12.2. An Incident_Station shall reference exactly one Station ID.

12.3. An Incident_Station shall store exactly one location note.

12.4. An Incident_Station shall store exactly one area type.

12.5. An Incident_Station shall store exactly one linked timestamp.

12.6. An Incident_Station (Incident_Report ID, station ID) pair shall be unique.

13. Incident_Train

13.1. An Incident_Train shall reference exactly one incident report ID.

13.2. An Incident_Train shall reference exactly one train ID.

13.3. An Incident_Train shall store exactly one train section note.

13.4. An Incident_Train shall store exactly one impact type.

13.5. An Incident_Train shall store exactly one linked timestamp.

13.6. An Incident_Train (incident report ID, train ID) pair shall be unique.

14. Incident_Photo

14.1. An Incident_Photo shall reference exactly one Incident_Report ID.

14.2. An Incident_Photo shall store exactly one photo URL.

14.3. An Incident_Photo shall store exactly one timestamp.

14.4. An Incident_Photo shall store zero or one caption.

14.5. A (Incident_Report ID, photo sequence number) pair shall be unique.

15. Incident_Status

15.1. An Incident_Status shall have exactly one unique status ID.

15.2. An Incident_Status shall have exactly one unique name.

15.3. An Incident_Status shall store exactly one description.

15.4. An Incident_Status shall store exactly one display order value.

15.5. An Incident_Status shall be assigned to zero or many Incident_Report.

16. Incident_History

16.1. An Incident_History shall have exactly one unique history ID.

16.2. An Incident_History shall reference exactly one incident report ID.

16.3. An Incident_History shall store exactly one timestamp.

16.4. An Incident_History shall store exactly one action description.

16.5. An Incident_History shall store exactly one action type.

17. Report_Category

17.1. A Report_Category shall have exactly one unique category ID.

17.2. A Report_Category shall have exactly one unique category name.

17.3. A Report_Category shall store exactly one description.

17.4. A Report_Category shall store exactly one escalation flag.

17.5. A Report_Category shall be assigned to zero or many Incident_Report.

18. Dispatch_Low_Level

18.1. A Dispatch_Low_Level shall have exactly one unique low level dispatch ID.

18.2. A Dispatch_Low_Level shall reference exactly one incident report ID.

18.3. A Dispatch_Low_Level shall reference zero or one employee ID.

18.4. A Dispatch_Low_Level shall reference zero or one officer ID.

18.5. A Dispatch_Low_Level shall store exactly one dispatched timestamp.

18.6. A Dispatch_Low_Level shall store exactly one dispatch status.

18.7. A Dispatch_Low_Level shall store zero or one response note.

18.8. A Dispatch_Low_Level shall have at least one of employee ID or officer ID.

19. Car

19.1. A car shall have exactly one unique car ID.

19.2. A car shall have exactly one unique car name.

19.3. A car shall belong to exactly one agency.

19.4. A car shall have exactly one car capacity.

19.5. A car shall have exactly one car type.

19.6. A car shall reference exactly one Vehicle_Status.

20. Train

20.1. A train shall have exactly one unique train ID.

20.2. A train shall have exactly one unique train code.

20.3. A train shall belong to exactly one agency.

20.4. A train shall have exactly one total capacity value.

20.5. A train shall have exactly one service type.

20.6. A train shall consist of one or many cars through Train_Car.

20.7. A train shall reference exactly one Vehicle_Status.

21. Train_Car

21.1. A Train_Car shall reference exactly one Train ID.

21.2. A Train_Car shall reference exactly one car ID.

21.3. A Train_Car shall store exactly one car order value.

21.4. A Train_Car shall store exactly one coupled timestamp.

21.5. A Train_Car shall store exactly one car role.

21.6. A Train_Car (train ID, car ID) pair shall be unique.

22. Vehicle_Status

22.1. A vehicle status shall have exactly one unique status ID.

22.2. A vehicle status shall have exactly one unique status name.

22.3. A vehicle status shall store exactly one description.

22.4. A vehicle status shall store exactly one operational flag.

22.5. A vehicle status shall be assigned to zero or many cars.

22.6. A vehicle status shall be assigned to zero or many trains.

23. Distance

23.1. A distance shall have exactly one unique distance ID.

23.2. A distance shall reference exactly one user ID.

23.3. A distance shall reference exactly one station ID.

23.4. A distance shall store exactly one value in miles.

23.5. A distance shall have exactly one calculated timestamp.

23.6. A distance shall store exactly one calculation method.

23.7. A (user ID, station ID) pair may appear multiple times.

24. Weather

24.1. A weather status shall have exactly one unique weather ID.

24.2. A weather status shall store exactly one weather status.

24.3. A weather status shall store exactly one temperature value.

24.4. A weather status shall store exactly one recorded timestamp.

24.5. A weather status shall reference exactly one data feed ID.

24.6. A weather shall be assigned to zero or many incident reports to indicate weather at the time of the incident.

25. Hazards

25.1. A hazard shall have exactly one unique hazard ID.

25.2. A hazard shall have exactly one unique name.

25.3. A hazard shall store exactly one description.

25.4. A hazard shall store exactly one severity class.

25.5. A hazard shall be assigned to zero or many Incident_Report to indicate associated hazards.

26. Incident_Hazard

26.1. An Incident_Hazard shall reference exactly one incident report ID.

26.2. An Incident_Hazard shall reference exactly one hazard ID.

26.3. An Incident_Hazard shall store exactly one hazard severity value.

26.4. An Incident_Hazard shall store exactly one detected timestamp.

26.5. An Incident_Hazard shall store exactly one mitigation status.

26.6. An Incident_Hazard (incident report ID, hazard ID) pair shall be unique.

27. Priority_Level

27.1. A Priority_Level shall have exactly one unique priority ID.

27.2. A Priority_Level shall have exactly one priority name.

27.3. A Priority_Level shall store exactly one description.

27.4. A Priority_Level shall store exactly one severity rank.

27.5. A Priority_Level shall be assigned to zero or many incident reports.

28. Saved_Station

28.1. A Saved_Station shall have exactly one unique saved station ID.

28.2. A Saved_Station shall reference exactly one user ID.

28.3. A Saved_Station shall reference exactly one station ID.

28.4. A Saved_Station shall have exactly one created timestamp.

28.5. A Saved_Station shall store zero or one custom label.

28.6. A Saved_Station shall store exactly one notification enabled flag.

28.7. A user shall have zero or many saved stations.

28.8. A (user ID, station ID) pair shall be unique.

29. Upvote

29.1. An upvote shall have exactly one unique upvote ID.

29.2. An upvote shall be created by exactly one user.

29.3. An upvote shall reference exactly one incident report.

29.4. An upvote shall have exactly one created timestamp.

29.5. An upvote shall store exactly one vote source.

29.6. An upvote shall store exactly one active flag.

29.7. A user shall create zero or many upvotes.

29.8. An incident report shall receive zero or many upvotes.

29.9. A (user ID, incident report ID) pair shall be unique.

30. Notification_Preference

30.1. A notification preference shall have exactly one unique preference ID.

30.2. A notification preference shall reference exactly one user.

30.3. A notification preference shall have exactly one notification type.

30.4. A notification preference shall have exactly one enabled flag.

30.5. A notification preference shall have exactly one delivery channel.

30.6. A user shall have zero or many notification preferences.

30.7. A (user ID, notification type) pair shall be unique.

31. Notification

31.1. A notification shall have exactly one unique notification ID.

31.2. A notification shall reference exactly one user.

31.3. A notification shall reference zero or one incident report.

31.4. A notification shall reference zero or one line.

31.5. A notification shall have exactly one created timestamp.

31.6. A notification shall have exactly one delivery status.

31.7. A notification shall reference exactly one notification type.

31.8. A user shall receive zero or many notifications.

32. Trip

32.1. A Trip shall have exactly one unique trip ID.

32.2. A Trip shall be operated by exactly one train.

32.3. A Trip shall follow exactly one line.

32.4. A Trip shall have exactly one scheduled start time.

32.5. A Trip shall have exactly one scheduled end time.

32.6. A Trip shall have exactly one trip status.

33. Stop

33.1. A Stop shall have exactly one unique stop ID.

33.2. A Stop shall reference exactly one trip ID.

33.3. A Stop shall reference exactly one station ID.

33.4. A Stop shall have exactly one scheduled arrival time.

33.5. A Stop shall have exactly one scheduled departure time.

33.6. A Stop shall have exactly one stop sequence number.

33.7. A (trip ID, station ID) pair shall be unique.

33.8. A (trip ID, stop sequence number) pair shall be unique.

34. Live_Feed

34.1. A Live_Feed shall have exactly one unique feed ID.

34.2. A Live_Feed shall reference exactly one stop ID.

34.3. A Live_Feed shall have exactly one update timestamp.

34.4. A Live_Feed shall store exactly one estimated arrival time.

34.5. A Live_Feed shall store exactly one current train capacity percentage.

35. Delay_Propagation

35.1. A Delay_Propagation shall have exactly one unique propagation ID.

35.2. A Delay_Propagation shall reference exactly one incident report ID.

35.3. A Delay_Propagation shall reference exactly one trip ID.

35.4. A Delay_Propagation shall reference exactly one stop ID.

35.5. A Delay_Propagation shall store exactly one calculated delay duration in minutes.

35.6. A Delay_Propagation shall store exactly one calculated timestamp.

35.7. A Delay_Propagation shall store exactly one impact scope.

36. Update

36.1. An Update shall have exactly one unique update ID.

36.2. An Update shall reference exactly one incident report ID.

36.3. An Update shall be created by exactly one user.

36.4. An Update shall have exactly one comment text.

36.5. An Update shall have exactly one created timestamp.

36.6. An Update shall have exactly one visibility status.

37. Vote

37.1. A Vote shall have exactly one unique vote ID.

37.2. A Vote shall be created by exactly one user.

37.3. A Vote shall reference exactly one update ID.

37.4. A Vote shall have exactly one vote type.

37.5. A Vote shall have exactly one created timestamp.

37.6. A Vote shall have exactly one active flag.

37.7. A (user ID, update ID) pair shall be unique.

38. Responder

38.1. A Responder shall have exactly one unique responder ID.

38.2. A Responder shall have exactly one unique responder name.

38.3. A Responder shall have exactly one description.

38.4. A Responder shall have exactly one responder category.

39. Dispatch_High_Level

39.1. A Dispatch_High_Level shall have exactly one unique high level dispatch ID.

39.2. A Dispatch_High_Level shall reference exactly one incident report ID.

39.3. A Dispatch_High_Level shall reference exactly one responder ID.

39.4. A Dispatch_High_Level shall reference exactly one station ID.

39.5. A Dispatch_High_Level shall store exactly one dispatched timestamp.

39.6. A Dispatch_High_Level shall store exactly one dispatch status.

39.7. A Dispatch_High_Level shall store zero or one priority note.

40. Fare_Log

40.1. A Fare_Log shall have exactly one unique log ID.

40.2. A Fare_Log shall reference exactly one incident report ID.

40.3. A Fare_Log shall reference exactly one user ID.

40.4. A Fare_Log shall reference exactly one station ID.

40.5. A Fare_Log shall have exactly one detected timestamp.

40.6. A Fare_Log shall store exactly one fare amount.

40.7. A Fare_Log shall store exactly one detection method.

41. Data_Feed

41.1. A Data_Feed shall have exactly one unique feed ID.

41.2. A Data_Feed shall have exactly one unique feed name.

41.3. A Data_Feed shall have exactly one feed type.

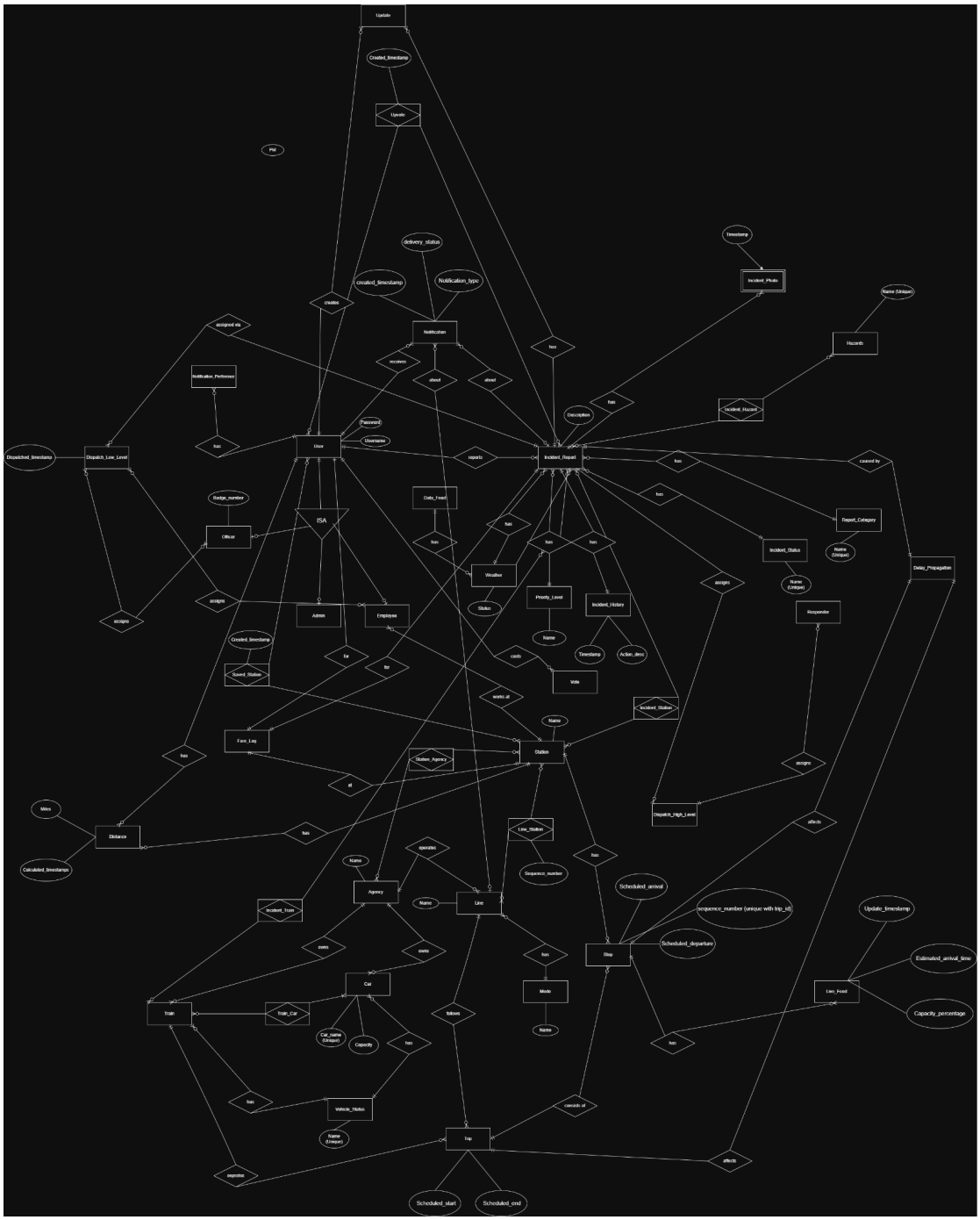
41.4. A Data_Feed shall have exactly one source URL.

Non-Functional Requirements

1. **Security & Privacy**
    - 1.1. Publicly visible reports shall hide the name of the reporter.
    - 1.2. “Criminal” reports shall be reviewed by staff until approved for the public feed.
    - 1.3. User passwords shall be encrypted in the database.
2. **Performance**
    - 2.1. Community portal shall support up to 5,000 concurrent users.
    - 2.2. Delay calculations shall complete within a few seconds.
    - 2.3. The system shall support 500 concurrent users during peak hours.
3. **Scalability**
    - 3.1. BART and Muni services are covered by our services.
    - 3.2. This database design will support adding new transit agencies without major changes in structure.
    - 3.3. The system shall handle 10,000 incidents without performance degradation.
4. **Reliability and Availability**
    - 4.1. Database backups should be performed daily.
    - 4.2. Database backups should be stored securely.
    - 4.3. The system shall have 100% uptime during operating hours.
5. **Storage**
    - 5.1. Incident reports shall be retained for at least 90 days.
    - 5.2. Each incident should support storage for up to 5 incident photos.
    - 5.3. Station and line data shall be pre-populated for BART and Muni.
6. **Privacy**
    - 6.1. Reporter name shall not appear in public view.
    - 6.2. User email addresses shall only be used only for notification.
    - 6.3. The database shall store only necessary user information.
7. **Maintainability and Coding Standards**
    - 7.1. Naming conventions shall be consistent across all tables.
    - 7.2. Each table should contain relevant comments describing its purpose.
    - 7.3. The database design shall be accompanied by an Enhanced Entity-Relationship Diagram.

- **Tradeoffs**:
    - By having public reports, the database will have more read requests and will be slower. However, we will limit report posting to one person at a time to prevent overflow of data.

Conceptual Design Enhanced Entity-Relationship Diagram



Entity Set Descriptions

1. **User**
   - 1.1. Primary Key(s): user_id: INT
   - 1.2. Foreign Key(s): None
   - 1.3. Attribute(s): username (unique, VARCHAR), password (VARCHAR), email (unique, VARCHAR)
   - 1.4. Description: Represents a system user who can submit incident reports and interact with the platform.

2. **Admin**
   - 2.1. Primary Key(s): admin_id: INT
   - 2.2. Foreign Key(s): user_id: INT (references User)
   - 2.3. Attribute(s): role_name (VARCHAR), access_level (VARCHAR), admin_status (VARCHAR)
   - 2.4. Description: System administrators who manage transit data including stations, lines, and agencies.

3. **Employee**
   - 3.1. Primary Key(s): employee_id
   - 3.2. Foreign Key(s): user_id (references User), station_id (references Station)
   - 3.3. Attribute(s): job_title (VARCHAR), employment_status (VARCHAR), hire_date (DATE)
   - 3.4. Description: Transit agency staff assigned to specific stations who can be dispatched to incidents.

4. **Officer**
   - 4.1. Primary Key(s): officer_id
   - 4.2. Foreign Key(s): user_id (references User)
   - 4.3. Attribute(s): badge_number (unique, VARCHAR), rank (VARCHAR), assigned_shift (VARCHAR)
   - 4.4. Description: Transit police or security personnel with badge numbers who can be dispatched to incidents.

5. **Agency**
    5.1. Primary Key(s): agency_id
    5.2. Foreign Key(s): None
    5.3. Attribute(s): name (unique, VARCHAR), headquarters_location (VARCHAR), contact_phone (VARCHAR)
    5.4. Description: Transit operators that own and operate lines, vehicles, and serve stations.

6. **Station**
    6.1. Primary Key(s): station_id
    6.2. Foreign Key(s): None
    6.3. Attribute(s): name (VARCHAR), city (VARCHAR), zone_code (VARCHAR)
    6.4. Description: Physical transit stops where passengers board and alight.

7. **Station_Agency**
    7.1. Primary Key(s): (station_id, agency_id) composite
    7.2. Foreign Key(s): station_id (references Station), agency_id (references Agency)
    7.3. Attribute(s): service_start_date (DATE), service_end_date (DATE), service_notes (VARCHAR)
    7.4. Description: Links stations to the agencies that serve them.

8. **Mode**
    8.1. Primary Key(s): mode_id
    8.2. Foreign Key(s): None
    8.3. Attribute(s): name (unique, VARCHAR), description (VARCHAR), typical_use_case (VARCHAR)
    8.4. Description: Types of transportation that lines can use (e.g., light rail, subway).

9. **Line**
    9.1. Primary Key(s): line_id
    9.2. Foreign Key(s): agency_id (references Agency), mode_id (references Mode)
    9.3. Attribute(s): name (unique, VARCHAR), line_color (VARCHAR), service_pattern (VARCHAR)

9.4. Description: Named transit routes that follow a specific sequence of stations.

10. Line_Station

10.1. Primary Key(s): (line_id, station_id) composite

10.2. Foreign Key(s): line_id (references Line), station_id (references Station)

10.3. Attribute(s): sequence_number (INT), platform_info (VARCHAR), direction_label (VARCHAR)

10.4. Description: Defines the ordered sequence of stations on a line.

11. Incident_Report

11.1. Primary Key(s): incident_id

11.2. Foreign Key(s): user_id (references User), status_id (references Incident_Status, optional), category_id (references Report_Category), weather_id (references Weather), priority_id (references Priority_Level)

11.3. Attribute(s): description (VARCHAR), reported_timestamp (TIMESTAMP), incident_source (VARCHAR)

11.4. Description: Core entity for all reported issues including accidents, hazards, maintenance needs, and safety concerns.

12. Incident_Station

12.1. Primary Key(s): (incident_id, station_id) composite

12.2. Foreign Key(s): incident_id (references Incident_Report), station_id (references Station)

12.3. Attribute(s): location_note (VARCHAR), area_type (VARCHAR), linked_timestamp (TIMESTAMP)

12.4. Description: Links incidents to the stations where they occurred.

13. Incident_Train

13.1. Primary Key(s): (incident_id, train_id) composite

13.2. Foreign Key(s): incident_id (references Incident_Report), train_id (references Train)

13.3. Attribute(s): train_section_note (VARCHAR), impact_type (VARCHAR), linked_timestamp (TIMESTAMP)

13.4. Description: Links incidents to specific trains involved.

14. Incident_Photo

14.1. Primary Key(s): None

14.2. Foreign Key(s): incident_id (references Incident_Report)

14.3. Attribute(s): photo_url (VARCHAR), timestamp (TIMESTAMP), caption (VARCHAR)

14.4. Description: Images uploaded by users to document incidents.

15. Incident_Status

15.1. Primary Key(s): status_id

15.2. Foreign Key(s): None

15.3. Attribute(s):name (unique, VARCHAR), description (VARCHAR), display_order (INT)

15.4. Description: Lookup table for incident lifecycles (e.g., 'Reported', 'In Progress', 'Resolved').

16. Incident_History

16.1. Primary Key(s): history_id

16.2. Foreign Key(s): incident_id (references Incident_Report)

16.3. Attribute(s): timestamp (TIMESTAMP), action_desc (TEXT), action_type (VARCHAR)

16.4. Description: Audit log of all actions taken on an incident.

17. Report_Category

17.1. Primary Key(s): category_id

17.2. Foreign Key(s): None

17.3. Attribute(s): name (unique, VARCHAR), description (VARCHAR), escalation_flag (BOOLEAN)

17.4. Description: Classification of incident types (Safety, Maintenance, Environmental).

18. Dispatch_Low_Level

18.1. Primary Key(s): low_level_dispatch_id

18.2. Foreign Key(s): incident_id (references Incident_Report), employee_id (references Employee, optional), officer_id (references Officer, optional)

18.3. Attribute(s): dispatched_timestamp (TIMESTAMP), dispatch_status (VARCHAR), response_notes (VARCHAR)

18.4. Description: Records which employees and officers are assigned to respond to incidents. At least one of employee or officer must be present.

19. Car

19.1. Primary Key(s): car_id

19.2. Foreign Key(s): agency_id (references Agency), vehicle_status_id (references Vehicle_Status)

19.3. Attribute(s): car_name (unique, VARCHAR), capacity (INT), car_type (VARCHAR)

19.4. Description: Individual railcars that can be coupled to form trains.

20. Train

20.1. Primary Key(s): train_id

20.2. Foreign Key(s): agency_id (references Agency), vehicle_status_id (references Vehicle_Status)

20.3. Attribute(s): train_code (unique, VARCHAR), capacity_total (INT), service_type (VARCHAR)

20.4. Description: Assembled consists of one or more railcars that operate as a unit.

21. Train_Car

21.1. Primary Key(s): (train_id, car_id) composite

21.2. Foreign Key(s): train_id (references Train), car_id (references Car)

21.3. Attribute(s): car_order (INT), coupled_timestamp (TIMESTAMP), car_role (VARCHAR)

21.4. Description: Links trains to their constituent cars over time.

22. Vehicle_Status

22.1. Primary Key(s): status_id

22.2. Foreign Key(s): None

22.3. Attribute(s): name (unique, VARCHAR), description (VARCHAR), operational_flag (BOOLEAN)

22.4. Description: Lookup table for vehicle operational status (Active, Maintenance, Out of Service).

23. Distance

23.1. Primary Key(s): distance_id

23.2. Foreign Key(s): user_id (references User), station_id (references Station)

23.3. Attribute(s): miles (DECIMAL), calculated_timestamp (TIMESTAMP), calculation_method (VARCHAR)

23.4. Description: Stores calculated distances between a user's location and a station at a specific time. Multiple records per user-station pair allowed.

24. Weather

24.1. Primary Key(s): weather_id

24.2. Foreign Key(s): data_feed_id (references Data_Feed)

24.3. Attribute(s): status (VARCHAR), temperature (DECIMAL), recorded_timestamp (TIMESTAMP)

24.4. Description: Weather conditions at the time of incidents, sourced from external meteorological networks.

25. Hazards

25.1. Primary Key(s): hazard_id

25.2. Foreign Key(s): None

25.3. Attribute(s): name (unique, VARCHAR), description (VARCHAR), severity_class (VARCHAR)

25.4. Description: Lookup table for specific danger types (e.g., spill, fire, downed power line).

26. Incident_Hazard

26.1. Primary Key(s): (incident_id, hazard_id) composite

26.2. Foreign Key(s): incident_id (references Incident_Report), hazard_id (references Hazards)

26.3. Attribute(s): hazard_severity (VARCHAR), detected_timestamp (TIMESTAMP), mitigation_status (VARCHAR)

26.4. Description: Links incidents to specific hazards present.

27. Priority_Level

27.1. Primary Key(s): priority_id

27.2. Foreign Key(s): None

27.3. Attribute(s): name (VARCHAR), description (VARCHAR), severity_rank (INT)

27.4. Description: Defines incident urgency (Low, Medium, High, Critical).

28. Saved_Station

28.1. Primary Key(s): saved_id

28.2. Foreign Key(s): user_id (references User), station_id (references Station)

28.3. Attribute(s): created_timestamp (TIMESTAMP), custom_label (VARCHAR), notification_enabled (BOOLEAN)

28.4. Unique Constraints: (user_id, station_id) unique

28.5. Description: User bookmarks or favorite stations for quick access.

29. Upvote

29.1. Primary Key(s): upvote_id

29.2. Foreign Key(s): user_id (references User), incident_id (references Incident_Report)

29.3. Attribute(s): created_timestamp (TIMESTAMP), vote_source (VARCHAR), active_flag (BOOLEAN)

29.4. Unique Constraints: (user_id, incident_id) unique

29.5. Description: Community validation mechanism allowing users to signal agreement with reported incidents.

30. Notification_Preference

30.1. Primary Key(s): pref_id

30.2. Foreign Key(s): user_id (references User)

30.3. Attribute(s): notification_type (VARCHAR), enabled_flag (BOOLEAN), delivery_channel (VARCHAR)

30.4. Unique Constraints: (user_id, notification_type) unique

30.5. Description: User-configurable settings for which alert types they wish to receive.

31. Notification

31.1. Primary Key(s): notification_id

31.2. Foreign Key(s): user_id (references User), incident_id (references Incident_Report, optional), line_id (references Line, optional)

31.3. Attribute(s): created_timestamp (TIMESTAMP), delivery_status (VARCHAR), notification_type (VARCHAR)

31.4. Description: Records of alerts sent to users about incidents, delays, or updates.

32. Trip

32.1. Primary Key(s): trip_id

32.2. Foreign Key(s): train_id (references Train), line_id (references Line)

32.3. Attribute(s): scheduled_start (TIMESTAMP), scheduled_end (TIMESTAMP), trip_status (VARCHAR)

32.4. Description: Scheduled journey of a specific train along a specific line at a specific time.

33. Stop

33.1. Primary Key(s): stop_id

33.2. Foreign Key(s): trip_id (references Trip), station_id (references Station)

33.3. Attribute(s): scheduled_arrival (TIMESTAMP), scheduled_departure (TIMESTAMP), sequence_number (INT)

33.4. Unique Constraints: (trip_id, station_id) unique, (trip_id, sequence_number) unique

33.5. Description: Individual scheduled station visits within a trip.

34. Live_Feed

34.1. Primary Key(s): feed_id

34.2. Foreign Key(s): stop_id (references Stop)

34.3. Attribute(s): update_timestamp (TIMESTAMP), estimated_arrival_time (TIMESTAMP), capacity_percentage (DECIMAL)

34.4. Description: Real-time updates to scheduled stops with current estimated arrival times and train capacity.

35. Delay_Propagation

35.1. Primary Key(s): propagation_id

35.2. Foreign Key(s): incident_id (references Incident_Report), trip_id (references Trip), stop_id (references Stop)

35.3. Attribute(s): delay_minutes (INT), calculated_timestamp (TIMESTAMP), impact_scope (VARCHAR)

35.4. Description: Tracks how a single incident causes cascading delays downstream. Records calculated impact on subsequent stops.

36. Update

36.1. Primary Key(s): update_id

36.2. Foreign Key(s): incident_id (references Incident_Report), user_id (references User)

36.3. Attribute(s): comment_text (TEXT), created_timestamp (TIMESTAMP), visibility_status (VARCHAR)

36.4. Description: Public comments and status updates posted by users about incidents for community transparency.

37. Vote

37.1. Primary Key(s): vote_id

37.2. Foreign Key(s): user_id (references User), update_id (references Update)

37.3. Attribute(s): vote_type (VARCHAR), created_timestamp (TIMESTAMP), active_flag (BOOLEAN)

37.4. Unique Constraints: (user_id, update_id) unique

37.5. Description: Community feedback on public updates to help surface accurate information.

38. Responder

38.1. Primary Key(s): responder_id

38.2. Foreign Key(s): None

38.3. Attribute(s): name (unique, VARCHAR), description (TEXT), responder_category (VARCHAR)

38.4. Description: Lookup table for first responder types (Police, Fire, EMS) for proper dispatch categorization.

39. Dispatch_High_Level

39.1. Primary Key(s): high_level_dispatch_id

39.2. Foreign Key(s): incident_id (references Incident_Report), responder_id (references Responder), station_id (references Station)

39.3. Attribute(s): dispatched_timestamp (TIMESTAMP), dispatch_status (VARCHAR), priority_note (VARCHAR)

39.4. Description: Records of first responder units sent to incidents at specific stations.

40. Fare_Log

40.1. Primary Key(s): log_id

40.2. Foreign Key(s): incident_id (references Incident_Report), user_id (references User), station_id (references Station)

40.3. Attribute(s): detected_timestamp (TIMESTAMP), fare_amount (DECIMAL), detection_method (VARCHAR)

40.4. Description: Automated records of fare evasion detection linked to low-priority incidents.

41. Data_Feed

41.1. Primary Key(s): feed_id

41.2. Foreign Key(s): None

41.3. Attribute(s): feed_name (unique, VARCHAR), feed_type (VARCHAR), source_url (VARCHAR)

41.4. Description: Metadata about external data sources integrated with the system (e.g., weather APIs, schedule feeds)